

1. Meeting Overview

Date: 2026-01-27

Time: 09:00 - 10:30 AM

Location: Conference Room B / Local H100 Cluster Node

Attendees: Alice (PI), Bob (Senior Researcher), Chloe (Machine Learning Engineer), Daniel (Junior Research Assistant)

2. Discussion Summary: Tau-Protein Data Gap

The primary focus of today's session was a critical data quality issue raised by **Bob**. During the ingestion of the longitudinal cohort, Bob identified a significant "Tau-protein" data gap affecting approximately 22% of the active subjects in the AD-Fusion-v2 pipeline.

2.1 Audit Directive

The discrepancy involves missing PET-Tau tracers for the 24-month follow-up window. **Alice** expressed significant concern regarding the potential for bias in our generative outputs if this missingness is non-random. She issued a stern instruction for **Bob** to halt current feature engineering and perform a comprehensive data audit. The goal is to determine if this is a systematic acquisition error or a failure in our internal ETL (Extract, Transform, Load) scripts.

Alice emphasized that we cannot proceed with the latent diffusion training until the source of this "missingness" is quantified and documented.

3. Model Progress: Architectural Adaptations

Despite the data gap, **Chloe** reported progress on the transformer backbone. To maintain momentum while Bob addresses the Tau-protein issue, she proposed an architectural workaround using dynamic masking.

3.1 Cross-Attention Layer Testing

Chloe plans to test a modified **Cross-Attention layer** designed to handle incomplete multimodal inputs. By implementing a learnable mask token for missing Tau-protein sequences, the model should, in theory, remain stable even when certain modalities are absent.

Chloe will begin benchmarking this layer on the current "incomplete" dataset to evaluate the attention weights. This will help us understand if the model is over-relying on T1-weighted MRI features in the absence of biochemical markers.

4. Pipeline Maintenance (Environment Status)

As documented in the last log (2026-01-18), the Dockerized environment on the **H100 cluster** is performing well. I have verified that Chloe's recent pushes to the dev branch do not conflict with the PyTorch 2.4/CUDA 12.1 stack.

However, to support the next phase of the project, I am preparing a new template based on the **OASIS-3** data structure to ensure our internal metadata remains standardized as we scale our subject count.

5. Action Items and Deadlines

Ref #	Task Description	Assigned To	Deadline
AI-01	Full Tau-Protein Audit: Document systematic vs. random missingness in PET-Tau tracers.	Bob	2026-01-30 (Friday)
AI-02	XAI Saliency Maps: Generate first-pass Explainable AI (XAI) maps for Alice's clinical review.	Chloe	2026-02-03
AI-03	Template Update: Push the new OASIS-3 standardized template to the shared GitLab repo.	Daniel	2026-01-29
AI-04	DPA Status Follow-up: Final check with Legal regarding the cloud processing ban.	Alice	2026-02-05

6. Closing Remarks

The team is currently at a critical junction. While the hardware and environment are optimized, the integrity of our biological priors (Tau-protein) is paramount. We will reconvene next Monday to review Bob's audit findings before committing further H100 compute hours to the full-scale training run.

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